

- ① What is the effect on threshold frequency and stopping potential on increasing the frequency of the incident beam of light? Justify your answer?
- ② The kinetic energies of a proton and that of an alpha particle are 4 eV and 1 eV, respectively. The ratio of the de Broglie wavelength associated with them will be :- a) 2:1 b) 1:1 c) 1:2 d) 4:1
- ③ An astronomical telescope may be a refracting type or reflecting type. Which of the two produces image of better quality? Justify your answer.
- ④ a) Explain the formation of energy bands in crystalline solids
b) Draw the energy band diagrams of (i) metal and (ii) semiconductor.
- ⑤ Differentiate between self inductance and mutual inductance.
- ⑥ Explain with the help of a diagram the working of a step down transformer. Why is a laminated iron core used in a transformer?
- ⑦ Two objects P and Q when placed at different positions in front of a concave mirror of focal length 20 cm, form real images of equal sizes. Size of object P is three times in size of object Q. If the distance of P is 50 cm from the mirror, then the distance of Q from the mirror is?